# ANATOMY OF AI AGENTS 2026

www.arianprabowo.com · 7-Layer OSI-Style Stack Reference (Based on Roitman arXiv:2606.24937)

**What actually runs under the hood of an autonomous AI agent in 2026?**
**Like the 7-layer OSI networking model, modern agents are a deep vertical stack — from GPU silicon (L1) & uv/Docker (L2) to RLVR weights (L4), MCP/Harness² (L5–L6) & A2A swarms (L7).**

① TRACE THE STACK: Follow Layers 1 → 7 from silicon to swarms
② BUILD EACH LAYER: Top 3 pills = flagship 2026 primitives & tools
③ HARDEN PROD: Audit the red FAILURE MODES & TRAPS on the right

*Synthesized from Haggai Roitman's 'The Hitchhiker's Guide to Agentic AI: From Foundations to Systems' (arXiv:2606.24937, v1.4, June 2026) + 2026 systems tooling. Companion to AI Skill Tree 2026 & AI Ladder Evolution by Arian Prabowo · Updated: 2026-11-07 19:00:11 AEDT (UTC+11:00)*

| OSI AGENT LAYER | CORE PRIMITIVES & ARCHITECTURE | FRAMEWORKS, PACKAGES & PROTOCOLS | SCALING, VERIFICATION & SECURITY | FAILURE MODES & TRAPS |
|---|---|---|---|---|
| OSI Analogy · PDU · Scope · Ch. | Hardware Physics · Math · Data Structures · Control Topologies | CLI Package Managers · DL Frameworks · Serving · Agent SDKs | Parallelism · Sandboxes · Evals & Oracles · Guardrails | What Breaks Here (Anti-Patterns) |
| **LAYER 1**<br>**PHYSICAL & SILICON**<br>Accelerators, memory hierarchies (SRAM vs. HBM3e), and high-speed interconnects executing tensor math.<br>• OSI L1 Physical · PDU: FLOPs & HBM Bytes<br>• Guide: Ch. 2.1, 11.10–14, 30.5 (GPU & Cluster Specs). | **GPU / TPU / NPU Tensor Cores**<br>**HBM3e vs. On-Chip SRAM Hierarchy**<br>**Arithmetic Intensity (FLOP/Byte)**<br>Streaming Multiprocessors (SMs) · Warp Schedulers (32-Thread SIMT)<br>Roofline Compute vs. Memory Bound · FP16 / BF16 / FP8 / FP4 Formats<br>Unified Memory Architecture (UMA) · High-Speed Die-to-Die Interconnect<br>PCIe Gen5/Gen6 Host-Device Bus · Wafer-Scale & SRAM-Only LPU Dies<br>Register File (256KB/SM) · L1 Cache / Shared Memory · L2 Cache (50MB–128MB)<br>HBM3e Bandwidth (3.3–8 TB/s) · SRAM Bandwidth (~20–30 TB/s) · Transformer Engine FP8/FP4<br>Stochastic Rounding · Tensor Memory Accelerator (TMA) · Systolic Array Matrix Multiply<br>Memory Coalescing · Bank Conflict Avoidance · TDP Power & Liquid Cooling | **NVIDIA H100 / H200 / Blackwell B200**<br>**Google TPU v5p / v6e Trillium / v7**<br>**Apple Silicon M4 Pro/Max/Ultra**<br>NVIDIA GB200 NVL72 Rack Scale · AMD Instinct MI300X / MI325X<br>AWS Trainium2 & Inferentia2 · Groq LPU & Cerebras WSE-3 ASIC<br>NVLink 4/5 (900–1,800 GB/s/GPU) · NVSwitch Intra-Rack Fabric<br>InfiniBand NDR/XDR (400–800Gb/s) · RoCEv2 Ethernet RDMA Fabric<br>DGX / HGX 8-GPU Node Topology · GPUDirect RDMA / Storage · Mellanox ConnectX-7 / 8 NICs<br>80GB / 141GB / 192GB VRAM Tiers · Mac Mini M4 64GB–512GB Cluster · RTX 4090 / 5090 Local Workstation<br>On-Device NPU (45–80+ TOPS) · Co-Packaged Silicon Photonics · Vera Rubin NVL144 (HBM4 Roadmap)<br>IPMI / Redfish BMC Telemetry · ECC Memory Error Correction · Gigawatt Data Center PPA | **Model FLOPs Utilization (MFU > 45%)**<br>**Prefill (Compute) vs. Decode (BW)**<br>**Confidential Compute GPU TEEs**<br>Prefill Phase (Matrix-Matrix Gemm) · Decode Phase (Matrix-Vector Gemv)<br>Multi-Instance GPU (MIG) Slices · Rail-Optimized Fat-Tree Topology<br>NVIDIA DCGM Hardware Health Check · Silent Data Corruption (SDC) Scans<br>Hot-Spare Node Auto-Failover · Hardware Root-of-Trust Attestation<br>MFU = Achieved / Peak FLOPs · Hardware FLOPs Utilization (HFU) · Bytes/Param: 2B (BF16) vs 0.5B (FP4)<br>4-Model RLHF VRAM Multiplier · Optimizer State Memory (12B/param) · Activation Memory Checkpointing<br>NVLink All-Reduce Bandwidth · Cross-Node All-to-All Bottleneck · Thermal Throttling Mitigation<br>H100/B200 Secure Enclave (CC-On) · Power Usage Effectiveness (PUE) · Straggler GPU Detection | **HBM Memory Bandwidth Wall (Decode)**<br>**CUDA Out-of-Memory (OOM) Crashes**<br>**Silent Tensor Core Bit-Flip Corrupt**<br>▸ Low Decode MFU (<5% Compute Util)<br>▸ PCIe Host-to-Device Bottleneck<br>▸ InfiniBand Link Flaps & Stalls<br>▸ Straggler GPU Halting 1,000 Nodes<br>• 4-Model RLHF VRAM Explosion<br>• Thermal Throttling Under RL Rollouts<br>• Unaligned Tensor Shapes (No TC)<br>• Consumer GPU P2P Disabled Locks |
| **LAYER 2**<br>**OS, ENV, PACKAGES & SANDBOX**<br>Operating systems, package managers (`Homebrew`, `uv`, `conda`, `pip`, `venv`), languages, and isolated sandboxes.<br>• OSI L2 Data Link · PDU: Syscalls, Wheels & MicroVMs<br>• Guide: Ch. 21.5, 22.1-9, 30.21 (Environments & Security). | **POSIX Processes, Threads & PTYs**<br>**Virtual Environments & Lockfiles**<br>**OS Namespaces, cgroups & Seccomp**<br>Python Interpreter & GIL / Free-Threaded · Asyncio Event Loop & Coroutines<br>ABI / manylinux Wheel Compatibility · Shared Libraries (.so / .dylib / DLL)<br>File Descriptors, Pipes (stdin/stdout) · Git Object Graph & Worktree Trees<br>Container UnionFS / OverlayFS Layers · Hardware Virtualization (KVM / HVF)<br>pyproject.toml (PEP 517 / 621) · uv.lock Deterministic Resolution · Content-Addressable Global Cache<br>Hardlink / Reflink Env Cloning · LD_LIBRARY_PATH / DYLD RPATH · fork() vs. spawn() Multiprocessing<br>IPC Shared Memory (/dev/shm) · POSIX Signals (SIGINT / SIGTERM) · eBPF Kernel Syscall Tracing<br>FUSE Virtual Filesystems · WebAssembly (WASI) Linear Memory · SSH / mTLS Socket Tunnels | **Homebrew (brew) · uv · conda / mamba**<br>**pip · venv · pipx · Poetry · Pixi**<br>**Python 3.12+ · Rust · TypeScript/Bun**<br>uv (uv venv / uv run / uvx / pip) · Conda / Miniforge / Mamba / Micromamba<br>Docker / Podman /OCI Containers · NVIDIA Container Toolkit (nvidia-docker)<br>Nix / Devcontainers / Direnv · Node.js / pnpm / npx (MCP Servers)<br>PyO3 & Maturin (Rust-in-Python) · Bash / Zsh / tmux / ripgrep / jq<br>brew install uv git ripgrep fd · uv pip compile & uv sync --locked · conda-lock & environment.yml<br>pip install --no-build-isolation · virtualenv & python -m venv .venv · Slurm / Kubernetes (K8s) / Ray<br>SQLite / DuckDB / PostgreSQL · Redis / Kafka Event Streams · GitHub Actions / GitLab CI Runners<br>CMake & Ninja Native Build · JupyterLab / Marimo Notebooks · Pydantic v2 (Rust Core Validation) | **E2B / Firecracker MicroVM Sandboxes**<br>**OS Sandboxing (Bubblewrap / Seatbelt)**<br>**OpenEnv Standardized Gym Reset/Step**<br>gVisor User-Space Kernel Isolation · Docker-in-Docker (DinD) Eval Farms<br>macOS sandbox-exec (Seatbelt Profiles) · Linux Bubblewrap (bwrap) Ro-Bind<br>Air-Gapped VLAN & Egress Firewall · Scoped Ephemeral OAuth / API Tokens<br>Immutable Read-Only Test Mounts · Snapshot & Sub-Second VM Restore<br>env.reset() / env.step(action) API · SWE-bench Docker Repo Images · OSWorld / WebArena Live Desktop VMs<br>Pyodide / Wasm Code Execution · Zero-Trust Network Policy (Cilium) · Secret Scrubbing in Env Vars<br>CPU / Memory / PID Fork-Bomb Quotas · Ephemeral Filesystem Wipe on Exit · SBOM & pip-audit Vulnerability Scan<br>Signed Container Attestation (Cosign) · Dedicated M4 Mac Mini Agent Host · Bastion Jump-Host Audit Logging | **Python Dependency & CUDA ABI Hell**<br>**Unsandboxed Host RCE & rm -rf /**<br>**Typosquatted PyPI / npm / Skill Packs**<br>▸ nvcc vs. torch.version.cuda Mismatch<br>▸ Global System Python Pollution<br>▸ pip install Breaking flash-attn ABI<br>▸ Agent Leaking .env & SSH Keys<br>• Container Escape via Privileged Flag<br>• Orphaned Zombie Processes & GPU Locks<br>• /dev/shm Exhaustion in PyTorch DDP<br>• GIL Contention Starving Async Loops |
| **LAYER 3**<br>**TENSOR, KERNELS & SERVING**<br>CUDA/Triton kernels, deep learning frameworks (`PyTorch`, `JAX`), distributed sharding, and LLM serving engines.<br>• OSI L3 Network · PDU: Autograd Graphs & Paged KV<br>• Guide: Ch. 1.6, 1.14-15, 2.2, 11.1-9 (Systems & vLLM). | **Autograd Computation Graphs & JIT**<br>**FlashAttention-2/3 IO-Aware Tiling**<br>**PagedAttention Virtual Memory KV**<br>Online Softmax Rescaling (Safe Max) · Continuous Iteration-Level Batching<br>Automatic Prefix KV Caching (Radix) · Speculative Decoding (Draft + Verify)<br>3D Parallelism (DP / TP / PP / SP) · ZeRO Stage 1 / 2 / 3 & FSDP Sharding<br>Expert Parallelism (EP) All-to-All · Disaggregated Prefill-Decode Routing<br>torch.compile (TorchInductor / Dynamo) · JAX jit / vmap / pmap / shard_map · XLA HLO Graph Fusion<br>CUDA Graphs (Zero Launch Overhead) · Ring Attention (Million-Token SP) · Medusa / EAGLE Multi-Head Drafting<br>1F1B & Interleaved Pipeline Schedule · Selective Activation Checkpointing · KV-Cache FP8 / INT4 Quantization<br>Chunked Prefill Scheduling · Guided Constrained Logit Masking · NCCL Ring & Tree All-Reduce | **PyTorch 2.x · JAX / Flax · MLX**<br>**CUDA · OpenAI Triton · FlashInfer**<br>**vLLM · SGLang · TensorRT-LLM · llama.cpp**<br>DeepSpeed ZeRO & Megatron-LM · PyTorch FSDP2 & DTensor<br>HuggingFace Transformers & Accelerate · SGLang RadixAttention & FSM Engine<br>llama.cpp / Ollama / LM Studio (GGUF) · bitsandbytes (NF4) · AWQ · GPTQ · EXL2<br>Outlines / XGrammar (JSON/CFG) · Ray Serve & NVIDIA Triton Server<br>cuBLAS / cuDNN / CUTLASS C++ · AMD ROCm / HIP Kernel Stack · Apple MLX Unified Array Engine<br>ONNX Runtime & TorchScript · HuggingFace Safetensors Zero-Copy · HuggingFace Tokenizers (Rust BPE)<br>vLLM AsyncLLMEngine & Paged KV · NVIDIA Dynamo Disaggregated Mesh · ExLlamaV2 / v3 Local Inference<br>BitNet.cpp 1.58-Bit Ternary Engine · Liger-Kernel Fused Triton Ops · NCCL / Gloo Collective Backend | **TTFT (Time-to-First-Token) & TPOT SLAs**<br>**Decoupled RL Rollout vs. Train Sync**<br>**Deterministic Constrained Decoding**<br>KV-Cache Eviction & Preemption Policy · NCCL Weight Broadcast (Trainer->vLLM)<br>PyTorch Profiler & Nsight Systems · Gradient Norm Clipping & NaN Guards<br>Distributed Async Checkpointing · Per-Tenant LoRA Adapter Hot-Swapping<br>Prefix-Cache Hit-Rate Optimization · Grammar-Constrained 100% Valid JSON<br>TTFT < 200ms · TPOT < 15ms/tok · Goodput (SLO-Compliant Tok/s) · RL Generation Bottleneck (70–90% Time)<br>In-Flight Weight Update (Zero Stall) · S-LoRA / Punica Multi-Adapter Batch · Torch Distributed Elastic Restart<br>NCCL_DEBUG & Flight Recorder · FP8 Delayed vs. Dynamic Scaling · Activation Outlier Channel Smoothing<br>Perplexity & KL Quantization Audit · Deterministic Seed Mode (cuDNN) · CUDA IPC Zero-Copy Weight Sharing | **KV-Cache Fragmentation & Preemption**<br>**RL Rollout Generation Bottleneck**<br>**Naive INT4 Quantization Collapse**<br>▸ 60–80% Contiguous KV Memory Waste<br>▸ Pipeline Parallel Bubble Idle Time<br>▸ torch.compile Graph-Break Thrashing<br>▸ NCCL Collective Deadlock / Timeout<br>• FP16 Loss-Scaling Underflow / NaNs<br>• Stale Policy Weights in Async RL<br>• Head-of-Line Blocking by Long Prompts<br>• CPU Tokenization / Detokenization Lag |
| **LAYER 4**<br>**FOUNDATION MODELS & RLVR**<br>Transformer/MoE/hybrid architectures, SFT, PEFT adapters, preference tuning, and verifiable-reward RL (`RLVR`).<br>• OSI L4 Transport · PDU: Token Logits & CoT Traces<br>• Guide: Ch. 1, 3–10, 12–13, 30.1–3 (Models, RL & Reasoning). | **Transformer · Sparse MoE · GQA / MLA**<br>**PEFT Adapters (LoRA / DoRA / PiSSA)**<br>**RLVR: PPO · DPO/KTO · GRPO / DAPO**<br>Pre-Norm RMSNorm & SwiGLU FFN · RoPE & YaRN Context Extension (1M+)<br>Hybrid Linear Attention (Gated DeltaNet) · Sequence Packing & Completion Masking<br>Bradley-Terry RM & Process RM (PRM) · GRPO Group-Relative Advantage Norm<br>DAPO Decoupled Clip & Dynamic Sample · Harness Distillation (Harness-Zero)<br>Multi-Head Latent Attention (MLA) · Aux-Loss-Free MoE Load Balancing · rsLoRA (alpha/sqrt(r)) · QLoRA NF4<br>Zeroth-Order MeZO Forward-Only Tune · Dr. GRPO (Unbiased Length Norm) · IPO · ORPO · SimPO · Online DPO<br>RLCR Calibrated Brier Score Reward · On-Policy Self-Distillation (OPSD/RLCSD) · Test-Time Compute (Best-of-N / MCTS)<br>Cold-Start Pure-RL Emergence (R1-Zero) · MergeKit (SLERP / TIES / DARE) · AdamW vs. Muon Matrix Orthogonal | **GPT-5.4 / o3 · Claude 4.6/5 · Gemini 3**<br>**DeepSeek-R1/V3.2 · Qwen 3 · Llama 4**<br>**HF TRL · PEFT · verl · OpenRLHF · Unsloth**<br>TRL (SFTTrainer / DPOTrainer / GRPO) · verl (HybridEngine RLHF/RLVR at Scale)<br>OpenRLHF (Ray + vLLM Distributed RL) · Unsloth (2x Speed Triton Backprop)<br>Axolotl & LLaMA-Factory YAML Suites · HuggingFace Alignment Handbook<br>Prime Intellect verifiers & OpenR1 · Torchtune & NeMo Aligner<br>DeepSeek-R1-Distill (7B–70B Weights) · Qwen3-235B-A22B & QwQ-32B · Claude Sonnet 4.5 / Opus 4.6 / 5.5<br>OpenAI GPT-5.4-Codex & o4-mini · Gemini 2.5 / 3.1 Pro & 3.8 Flash · HuggingFace Datasets & Hub<br>FineWeb-Edu & Dolma Corpora · Lean 4 / OpenMathReasoning Verifiers · MergeKit CLI Checkpoint Merger<br>WandB / TensorBoard / MLflow Logs · vLLM Rollout Worker Integration · ChatML & Special Tool-Call Tokens | **Verifiable Execution Oracles (RLVR)**<br>**KL-Divergence & Entropy Regularization**<br>**Constitutional AI & Safety Classifiers**<br>Deterministic Unit-Test / Math Graders · Completion-Only Loss Masking (Labels=-100)<br>Multi-Objective Reward Combination · Listwise Rank-Based Reward Models<br>Instruction Hierarchy Privilege Training · Deliberative Alignment in CoT Traces<br>Activation Steering & SAE Circuit Probes · GCG / AutoDAN / Best-of-N Red-Teaming<br>KL Penalty beta * D_KL(pi_theta \|\| pi_ref) · PPO Clipping epsilon in [0.1, 0.28] · GRPO Group Size G = 8..64 Rollouts<br>Pass@k vs. Majority@k Scaling · Length-Controlled Win-Rate Audits · Catastrophic Forgetting Evals<br>Sleeper Agent Backdoor Detection · Sandbagging & Alignment Faking Audits · Machine Unlearning (RMU / Gradient Ascent)<br>CBRN / Cyber ASL-3/4 Threshold Gates · Contamination-Free LiveBench / HLE · Calibration Expected Error (ECE) | **Reward Hacking & Length Inflation**<br>**Entropy Collapse & Mode Collapse**<br>**Catastrophic Forgetting / Alignment Tax**<br>▸ GRPO Zero-Variance Advantage (All 0/1)<br>▸ DPO Likelihood Displacement (Both Drop)<br>▸ Training on Prompt Tokens (No Masking)<br>▸ Chat Template EOS Token Mismatch<br>• Sycophancy & Delusional Validation<br>• Emergent Misalignment from Narrow SFT<br>• Over-Refusal on Benign Triggers<br>• Model Collapse on Unfiltered Synthetic |
| **LAYER 5**<br>**CONTEXT, RAG, MEMORY & MCP**<br>Context window budgeting, Agentic RAG, multi-tier memory systems, `Model Context Protocol (MCP)`, and `SKILL.md`.<br>• OSI L5 Session · PDU: Context Windows & JSON-RPC<br>• Guide: Ch. 16, 17, 18.2-4, 23, 24 (RAG, Memory, MCP & Skills). | **Context Engineering & Token Budgeting**<br>**Hybrid & Agentic RAG (Dense+BM25+Graph)**<br>**4-Tier Memory + Sleep-Time Dreaming**<br>Working Memory (Active Context Window) · Episodic Memory (Past Experience Logs)<br>Semantic Memory (Entity/Fact Triples) · Procedural Memory (SKILL.md / Weights)<br>Reciprocal Rank Fusion (RRF) & Rerank · GraphRAG Community Entity Summaries<br>REFRAG Compress-Sense-Expand Decoding · Progressive Disclosure Skill Loading<br>System / History / RAG / Tool Scratchpad · Bi-Encoder (HNSW) vs. Cross-Encoder · Contextual Chunk Headers (Anthropic)<br>Late Interaction ColBERT Token MaxSim · Search-R1 RL-Trained Query Reformulation · Write / Read / Reflect / Consolidate Ops<br>Ebbinghaus Temporal Memory Decay · Sleep-Time Offline Consolidation (Dream) · Proactive Context Pre-Fetching<br>KV-Cache Prefix Pinning (90% Discount) · Observation Masking & Summary Compaction · JSON-RPC 2.0 Host-Client-Server Bus | **Model Context Protocol (MCP / .mcp.json)**<br>**Agent Skills (SKILL.md Folder Standard)**<br>**Qdrant · Milvus · pgvector · Neo4j**<br>MCP Primitives: Tools, Resources, Prompts · MCP Transports: stdio & Streamable HTTP<br>FastMCP Python / TypeScript SDKs · Letta (MemGPT) · Mem0 · Zep Memory<br>LlamaIndex & Haystack RAG Pipelines · BGE-M3 / Cohere Embed v3 & Rerank v3<br>Docling / Unstructured PDF Table Parsers · Linux Foundation AAIF MCP Registry<br>SKILL.md YAML Frontmatter + Instructions · skills/scripts/ & skills/references/ · MCP Sampling & Roots Client Callbacks<br>ChromaDB · Weaviate · Pinecone · LanceDB · Elasticsearch / OpenSearch BM25 · Microsoft GraphRAG & LightRAG<br>LangMem Semantic/Episodic Store · Cognee & Memary Knowledge Graphs · Firecrawl / Jina Reader Markdown Scrape<br>Tree-sitter AST Codebase Indexing · OpenAPI-to-MCP Auto-Generators · Anthropic & Cursor MCP Marketplaces | **RAG Triad Evals (Faithfulness / Recall)**<br>**MCP OAuth 2.1 Scopes & Tool Signing**<br>**/compact Hygiene & Context Pruning**<br>Ragas & TruLens Groundedness Oracles · NDCG@k · MRR · Recall@k Retrieval Audits<br>Document-Level RBAC / ACL Pre-Filtering · Indirect Prompt Injection Sanitization<br>Human-in-the-Loop Destructive Tool Gate · Tool Schema Token-Bloat Minimization<br>Memory Contradiction & Stale-Fact Purge · Citation Span-Locking & Provenance Check<br>Context Utilization < 65% Rot Threshold · Lost-in-the-Middle U-Curve Placement · Needle-in-a-Haystack (NIAH) & Ruler Eval<br>LOCOMO & LongMemEval Memory Benchmarks · MCP Server Code-Signing Verification · Least-Privilege Read-Only DB Connectors<br>PII Redaction Before Vector Indexing · Hallucination Span Detection (LettuceDetect) · Dynamic Tool Retrieval (Top-K Tools)<br>Episodic Replay Buffer Deduplication · Cross-Tenant Memory Isolation · OX Security stdio RCE Hardening | **Context Rot & Lost-in-the-Middle**<br>**Indirect Prompt Injection via RAG/MCP**<br>**MCP Tool Bloat & Malicious Skills**<br>▸ 100+ MCP Tool Schemas Blowing Context<br>▸ ClawHavoc Malicious SKILL.md Payloads<br>▸ MCP Rug-Pull & Tool Shadowing Attacks<br>▸ Naive Fixed 512-Token Mid-Table Splits<br>• Unfiltered Vector RAG Leaking Exec Docs<br>• Memory Poisoning Persisting Across Sessions<br>• Stale Episodic Memory Overriding New Code<br>• Hallucinated Citations on Unretrieved Facts |
| **LAYER 6**<br>**AGENT HARNESS, LOOP & GRAPH**<br>The CPU/OS of an agent: `CLAUDE.md`/`AGENTS.md`, Loop Engineering, Graph Engineering (`LangGraph`), and `Harness²`.<br>• OSI L6 Presentation · PDU: State Transitions & Diffs<br>• Guide: Ch. 18, 19, 20, 21, 27 (Harness, Loops & Graphs). | **Agent Harness (Prompt+Tools+State+Loop)**<br>**Loop Engineering (Observe→Decide→Act)**<br>**Graph Engineering (Typed StateGraphs)**<br>ReAct · Reflexion · Plan-and-Execute · 5 Loop Primitives (State/Step/Check/Stop)<br>Typed State Schema & Reducer Functions · Conditional Edges & Cyclic Graph Routing<br>Fan-Out / Fan-In Parallel Send() API · Recursive Agent Harnessing (Harness²)<br>Spec-Driven Development (Spec→Plan→Task) · Cognitive Escalation & Uncertainty Handoff<br>pi(a_t \| s_t) Inference-Time Policy Loop · LATS (Language Agent Tree Search) · LLM Compiler Parallel DAG Execution<br>BDI (Belief-Desire-Intention) Hybrid · Checkpointer Time-Travel & Replay · Human-in-the-Loop Interrupt() Gates<br>Stagnation & Repeat-Action Detection · Subagent Context Isolation (Forked Call) · Red-Green-Refactor TDD Agent Cycle<br>Exponential Backoff & Schema Self-Repair · Token/Cost/Iteration Termination Caps · Dynamic Model Routing (Fast vs. Reasoner) | **Claude Code CLI · Codex CLI · OpenCode**<br>**Cursor · Windsurf · Cline / Roo · Aider**<br>**LangGraph · OpenAI Agents SDK · DSPy**<br>CLAUDE.md & AGENTS.md Repo Constitutions · .cursor/rules & .github/copilot-instructions<br>GitHub Spec Kit & obra/superpowers TDD · LlamaIndex Workflows (Event-Driven)<br>Smolagents (CodeAct Python-as-Action) · PydanticAI Type-Safe Agent Harness<br>OpenHands (OpenDevin) & Block goose · Semantic Kernel & Haystack Agents<br>LangGraph PostgresCheckpointer · DSPy MIPROv2 Prompt/Weight Compiler · Claude Agent SDK & Computer Use API<br>Browser-Use & Playwright DOM Harnesses · Vercel AI SDK Core (generateText/maxSteps) · Guardrails AI & NeMo Guardrails<br>Instructor & BAML Structured Outputs · Litellm Unified Model Router · Git Pre-Commit & CI Harness Hooks<br>Repomix / Gitingest Codebase Packers · LangSmith · Arize Phoenix · Langfuse · OpenTelemetry (OTel) GenAI Spans | **Decoupled Verification Oracles (No Self-Grade)**<br>**SWE-bench Verified/Pro & Terminal-Bench**<br>**pass^k Reliability & Trajectory Tracing**<br>Property-Based Testing (Hypothesis/Kani) · Mutation Testing (Stryker / mutmut) Gates<br>Immutable CI Tests Locked From Agent Write · Trajectory Tool-Call Exact/Prefix Match<br>LLM-as-a-Judge Pairwise Rubric Grading · Cost-per-Resolved-Task Efficiency Metric<br>PyRIT & Garak Adversarial Red-Teaming · LlamaFirewall Prompt/Tool Guardrails<br>pass^k = (pass@1)^k Consistency Metric · METR 50%-Reliability Horizon (2h-30h) · GAIA · OSWorld · WebArena · TAU-bench<br>Scale SWE-bench Pro (Private Repos) · Step Efficiency (Optimal / Actual Steps) · Deterministic Mock-Tool Replay Harness<br>Shadow-Mode Traffic Counterfactual Eval · Circuit-Breaker Spend & Loop-Depth Caps · Canary Token Exfiltration Tripwires<br>Git Diff AST Security Linter (Semgrep) · Automated Rollback on Regression · EU AI Act / SOC2 Immutable Audit Logs | **Agent Deleting / Mocking Failing Tests**<br>**Unbounded Loop Burn & Action Stagnation**<br>**God-Node Anti-Pattern & State Bloat**<br>▸ Coupled Agent Rewriting Its Own Grader<br>▸ 0.95^20 = 35.8% Compounding Step Decay<br>▸ Poisoned .cursor/rules or GitHub Issues<br>▸ CVE-2025-53773 IDE Config Auto-Approve RCE<br>• Swallowing Exceptions in Tool Wrappers<br>• Appending Raw 50KB HTML to Graph State<br>• Premature Termination Before Verification<br>• Sycophantic Plan Pivot on User Pushback |
| **LAYER 7**<br>**MULTI-AGENT MESH, A2A & UI**<br>Multi-agent topologies (`OWC`), inter-agent protocols (`A2A` / `ACP`), parallel worktree swarms, and Agentic/Generative UI.<br>• OSI L7 Application · PDU: A2A Tasks & GenUI Streams<br>• Guide: Ch. 21.6, 25, 26, 28, 30.10/18 (A2A, MAS & Agentic UI). | **Orchestrator-Worker-Critic (OWC) Pattern**<br>**Multi-Agent Topologies (Hierarchical/Mesh)**<br>**Generative UI & Shared State Sync**<br>Centralized Supervisor vs. Decentralized · Hierarchical Tree (Squads of Squads)<br>Debate / Society-of-Mind Consensus · Blackboard Shared-Memory Coordination<br>Contract-Net Task Bidding & Auctions · Parallel Git-Worktree Swarm Isolation<br>A2A (Horizontal) × MCP (Vertical) Dual Bus · Human-on-the-Loop Escalation & Approval<br>Agent Card (/.well-known/agent.json) · Task Lifecycle (Submitted→Working→Done) · Multi-Agent MDP & CTDE MARL Training<br>Credit Assignment (Shapley / Counterfactual) · Role Specialization & Persona Diversity · Byzantine Fault-Tolerant Critic Voting<br>Pub/Sub Asynchronous Event Choreography · Process-Centric Enterprise State Workflows · Server-Driven UI Component Streaming<br>Optimistic UI Updates & Diff Previews · Ambient OS Cross-App Delegation · Organizational Digital-Twin Simulation | **Google A2A Protocol · IBM ACP · ANP**<br>**CrewAI · AutoGen / AG2 · LangGraph Swarm**<br>**CopilotKit · AG-UI · Vercel AI SDK UI**<br>Google Agent-to-Agent (A2A) JSON-RPC/SSE · IBM BeeAI & Agent Communication Protocol<br>Microsoft AutoGen v0.4 / Semantic Kernel · CrewAI Flows & Role-Based Crews<br>Claude Code Subagents & Worktree Swarms · OpenClaw 24/7 Multi-Channel Agent Daemon<br>Agent Gateways (Kong / Solo / LiteLLM) · Claude Artifacts · ChatGPT/Gemini Canvas<br>AG-UI (Agent-User Interaction Protocol) · Vercel AI SDK useChat / streamUI Hooks · Chainlit · Streamlit · Gradio Agent UIs<br>OpenAI ChatKit & Realtime WebRTC Voice · LangGraph Platform Cloud & Studio UI · Camel-AI Workforce & MetaGPT SOPs<br>OpenAI Swarm / Agents SDK Handoffs · A2A Artifact & Push-Notification Webhooks · React Server Components (RSC) GenUI<br>Slack / Discord / WhatsApp Bot Bridges · Linear / GitHub PR Async Agent Teammates · 1-Click MCP & A2A Enterprise Registries | **Agent Gateway (mTLS, Auth, FinOps Caps)**<br>**Cross-Agent Trust & Cryptographic KYA**<br>**Independent Critic & Red-vs-Blue Swarms**<br>Zero-Trust Agent Identity (SPIFFE / KYA) · OAuth 2.1 On-Behalf-Of (OBO) Token Flow<br>Per-Swarm Token & Dollar Circuit Breakers · Anti-Collusion & Decorrelated Critic Models<br>Cross-Repo API Contract Locks (OpenAPI) · Deadlock, Livelock & Cycle Detection<br>Provenance Watermarking (C2PA / SynthID) · Strict Liability & Action Audit Ledgers<br>Communication Overhead O(N^2) Pruning · Multi-Agent Judge Independence Check · Cross-Org Data-Loss Prevention (DLP)<br>Swarm Amplification Factor Monitoring · Canary Deployment of Agent Topologies · Chaos Engineering (Injecting Rogue Agents)<br>XBOW / Cybench / T3MP3ST Cyber Ranges · Lean 4 / TLA+ Formal Protocol Verification · Human Escalation SLA Latency Tracking<br>Anti-Sycophancy & Anti-Psychosis Filters · GDPR / HIPAA Cross-Agent Consent Scopes · Kill-Switch Broadcast Across Swarm Nodes | **Cascading Hallucination & Swarm Echo**<br>**O(N²) Chatty Agent Token Explosion**<br>**Cross-Agent Confused Deputy & Escalation**<br>▸ 7-Persona Roleplay Chatter Burning 15x Cost<br>▸ Homogeneous Critic Blindly Agreeing<br>▸ Supervisor Bottleneck & Context Overflow<br>▸ Merge Conflicts Across Parallel Worktrees<br>• Rogue Subagent Bypassing Parent Scopes<br>• Infinite Ping-Pong Handoff Deadlocks<br>• Free-Riding Worker Agents in Crews<br>• UI Stream Desync & Phantom Tool States |